Market-Implied vs News-Based Uncertainty: Predicting Sector ETF Volatility

1. Introduction

Financial markets are not only affected by market fundamentals or macroeconomic conditions, but also by how investors interpret uncertainty. In response to periods of financial stress, policy concerns, geopolitical conflicts, risk expectations change, and along with it, market volatility. The midterm project explored the relationship between Economic Policy Uncertainty (EPU) and sector ETF volatility using monthly data. The preliminary analysis suggested high volatility was associated with high uncertainty periods, and the association was stronger for the Energy and Financial sectors, while less so for sectors like Healthcare. However, the midterm analysis was exploratory and used monthly data, which limited the ability to capture fast market reactions.

The final project extends the midterm in three ways. Firstly, the frequency of the data used in our analysis moves from monthly to daily data. By improving the time resolution, we hope to be able to better capture the fast reaction of the market to uncertainty. Secondly, the final project aims to forecast, and ask: Can uncertainty signals predict future sector ETF volatility? Thirdly, the final project compares different sources/forms of uncertainty signals. VIX captures market-implied expected volatility as it is a signal extracted from S&P500 option prices. The EPU in contrast, is a broad news based measure of uncertainty regarding economic policy based on U.S. newspapers. The Equity Market related Economic Uncertainty index (EMEU) is also news based, but instead of policy it is extracted from news specifically related to equity markets, which provides an alternative perspective of news based uncertainty.

The main research question is: **Do uncertainty signals improve out of sample ETF volatility forecasting beyond past market behavior?** This question is further split into four sub-questions: **1.** Which type of uncertainty signal provides the most predictive value? We compare three uncertainty signal families: VIX, EPU, and EMEU. Each signal family is compared against a baseline model which uses only past and recent market behaviour. **2.** Does the predictive value of uncertainty signals depend on model architecture? We apply the same feature set comparisons using three modelling architectures: Linear Regression, Random Forest, and XGBoost. This allows our analysis to explore whether certain uncertainty signals are only useful when modelled with more complex relationships. **3.** Does the predictive value of uncertainty signals depend on forecast horizon? We fit the same models once for the 5 day volatility response and once for the 21 day volatility response. This allows us to explore whether certain uncertainty signals are only useful in shorter or longer term forecasting horizons. **4.** Do uncertainty signals matter differently across sectors? We evaluate model performance separately for the 6 sector ETFs to examine whether uncertainty signals are more useful for certain sectors.

We hypothesize that past volatility will be the strongest predictor since volatility tends to persist over time. We expect uncertainty signals to provide additional value, especially VIX, as it is an uncertainty signal directly extracted from the market through option prices. We expect news based measures like EPU and EMEU to be less consistently useful, as their effects may be overshadowed by VIX since news based uncertainty may be reflected in options prices already. However, we do expect their effect to be more pronounced in the Energy and Financial sector which are more policy sensitive. We expect the usefulness of uncertainty signals to depend on model structure, especially since the relationship between uncertainty signals and realized volatility may be complex and not simply linear. Finally, we expect uncertainty signals to be more effective in the shorter horizon, as markets absorb uncertainty quickly.

2. Methods

2.1 Data

We use two datasets, daily ETF price data and daily uncertainty measures. The ETF data is collected on the same 6 ETFs used in the midterm, in order to have sufficient market and sector representation: SPY for the overall U.S. equity market, XLE for energy, XLF for financials, XLI for industrials, XLK for technology, and XLV for healthcare. The final dataset uses daily observations from 2006 to 2026 with each row representing one ETF on one trading day.

Table 1. Data sources and roles

Data	Variables	Source	Purpose
ETF prices	Adjusted close, open, high, low, close, volume	Yahoo Finance	Compute returns, realized volatility, volume features
Market implied uncertainty	VIX	FRED	Option market expectations of near term volatility
Policy news uncertainty	EPU	FRED	News based uncertainty related to economic policy
Equity market news uncertainty	EMEU	FRED	News based uncertainty related to equity markets

The final dataset is daily, rather than monthly like in the midterm. This is important because markets may price in uncertainty very quickly, and a monthly model may fall short on the quick timing of volatility reactions. A daily model allows the project to compare both short and long horizon predictability.

All data processing and modeling were conducted in Python. The packages used for data cleaning, merging and feature construction include pandas and numpy. Interactive visualisations were created using plotly.

2.2 Forecasting setup

This project is formulated as a supervised regression task. For each ETF on each trading day t, models only use information available up to that day to predict volatility from day $t + 1$ onward. This setup mitigates look ahead bias. While same day market reactions are definitely important, this project focuses on forecasting future volatility rather than explaining same day market movement.

Daily log return is defined as $r_t = log(P_t) - log(P_{t-1})$, where P_t is the adjusted closing price on day t. The response variables are 5 and 21 day volatility measures, defined respectively as $future\_vol\_5d_t = SD(r_{t+1}, ..., r_{t+5})$ and $future\_vol\_21d_t = SD(r_{t+1}, ..., r_{t+21})$. The choice of

5 and 21 days to represent short term and long term horizons is because 5 days represents one trading week, while 21 days represents approximately one trading month. Using both responses allows our analysis to additionally compare the predictive value of uncertainty signals in the short and long term.

2.3 Feature engineering

Rather than looking at each predictor as isolated variables, predictors are organised into feature families. This aligns with the goal of this project as we aren't investigating which transformation of EPU or VIX is best, but whether different sources of sentiment information improves volatility forecasting. The feature families are Market history, VIX, EPU, EMEU. Ticker indicators are also included to control for baseline sector differences.

Market history features

These features form the baseline feature set since financial volatility is commonly modelled using past information. These features include absolute return, past volatility, past return (cumulative), and volume changes. Same day absolute return captures the most recent price movement. 5 and 21 day past volatility captures short run volatility changes and long run volatility conditions. 5 and 21 day past cumulative returns capture information about market stress or momentum. 1 and 5 day volume change capture information on shifts in trading activity.

Uncertainty signal features

We engineer similar uncertainty signal features for each of our 3 different sources of uncertainty information. This is done so we can compare the signal families fairly. Raw level of each uncertainty index captures same day uncertainty measurement. The 5 day change captures recent shifts of uncertainty. The 21 day average captures the most recent monthly uncertainty conditions. The 252 day z score measures whether or not current uncertainty level is unusually high or low, using the previous trading year as reference. Finally, the 95th percentile spike indicator identifies states of extreme uncertainty relative to the past year.

Table 2. Feature families summary

Feature family	Features
Market history	Absolute return (Same day), past volatility (5 and 21 day), past return (5 and 21 day), volume change (1 and 5 day)
VIX	Level, change(5 day), average (21 day), z-score, spike
EPU	Level, changes, averages, z-score, spike
EMEU	Level, changes, averages, z-score, spike
Ticker indicator	SPY, XLE, XLF, XLI, XLK, XLV

These engineered features are intentionally interpreted at the feature-family level rather than as independent causal variables. We aim to represent each source of uncertainty signal across different meaningful time scales of immediate level, weekly change, monthly conditions and yearly abnormalities. This design choice means that many of the engineered features within the same family will be correlated, meaning we can’t reliably interpret individual coefficients. However, since the goal is to comparatively evaluate predictive performance of feature families rather than interpret causal coefficients, we accept this tradeoff.

2.4 Models

The same feature set comparisons were applied across three model architectures. Linear Regression serves as a benchmark model that allows us to investigate linear relationships between uncertainty signals and volatility. Random Forest is a nonlinear ensemble model that allows us to capture interactions and threshold behavior. Our RF model used 400 trees, a maximum tree depth of 8, a minimum samples per leaf of 10, and a fixed random seed of 370. These choices limit tree complexity while still allowing us to capture meaningful nonlinear interactions. XGBoost is a boosted tree model that also captures nonlinear patterns and interactions. We used 500 boosting rounds, a maximum depth of 3, learning rate of 0.03, subsampling ratio and column subsampling ratio of 0.8, a squared error regression objective, RMSE as the evaluation metric, and also a fixed seed of 370. These choices provide decent flexibility while reducing overfitting. This design choice aims to separate the questions of whether uncertainty signals improve volatility forecasting, and whether the effects depend on the modelling structure.

For each model class, we fit the same eight feature groups.

Table 3. Feature groups

B: Market history features, V: VIX features, P: EPU features, E: EMEU features.

B	$B + V$	$B + P$	$B + E$
$B + V + P$	$B + V + E$	$B + P + E$	$B + V + P + E$

With 2 target variables, 3 model classes, and 8 feature groups, we get a total of 48 model fits. This allows us to explore whether each uncertainty family (And combinations of them) improve test performance beyond past market behaviour. For any fixed feature group, Linear Regression, Random Forest, and XGBoost can also be compared to explore if nonlinear modelling structure improves prediction. Furthermore, we can also compare whether the feature groups perform better in short vs long horizon volatility forecasting.

2.5 Train/test design and evaluation metrics

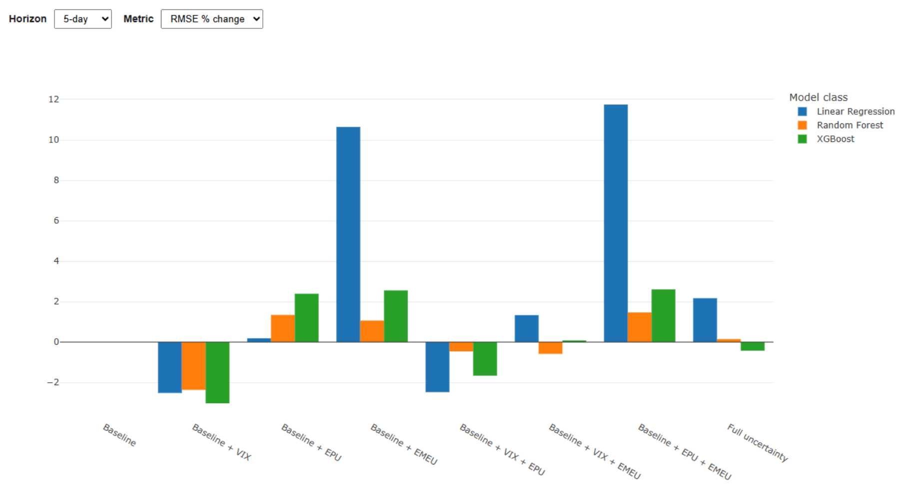
A time-based split is used rather than a random one. The training period covers 2006–2021, and the test period covers 2022–2026. This prevents future market data from leaking into training, reflecting a more realistic forecasting setting.

The primary evaluation metric is test RMSE, as it penalises larger forecasting errors more heavily. The secondary evaluation metric is test MAE which measures average absolute forecasting error. Test R^2 is used to summarise the proportion of variation explained in the test period. For each model class, we determine feature family (And combinations of them) improvements relative to the same model’s baseline (Only past market behaviour). This ensures that comparisons are made with the same modelling architecture.

Finally, model performance is also evaluated separately by ETF. The models are trained on a pooled dataset, but we summarise test errors within each ETF. This allows us to explore whether the effects of certain uncertainty signals differ by sector. We also use RF feature importance as a diagnostic tool, where we aggregate importance scores by feature families.

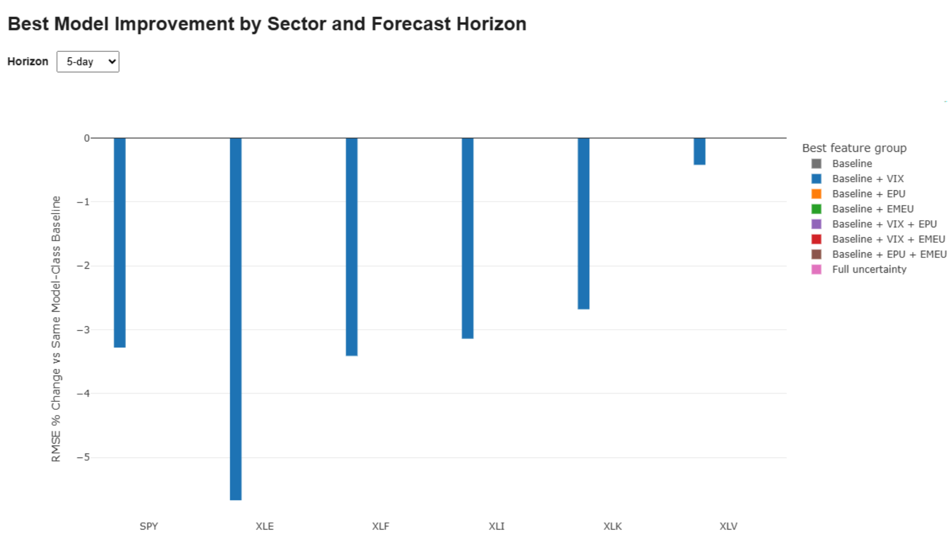
3. Results

Figure 1 Model Improvement Explorer (Interactive)



The results show that the predictive value of uncertainty signals depends strongly on forecasting horizon. For the 5 day target, VIX is the most consistent uncertainty signal. Adding VIX improves test RMSE of LR, RF and XGB by 2.52%, 2.37% and 3.03% respectively. However, adding EPU or EMEU generally worsens forecasting. For the 21 day target (See interactive version), results vary by model. For LR, adding VIX only improves RMSE by 0.05%. For RF, adding EPU and EMEU improves RMSE by 7.37%. For XGB, adding VIX and EPU improves RMSE by 3.47%. This suggests that in the short horizon, volatility forecasting benefits more from market implied uncertainty, while news based uncertainty have long term predictive value but only when non-linear models are used.

Figure 2 Sector Improvement Explorer (Interactive)

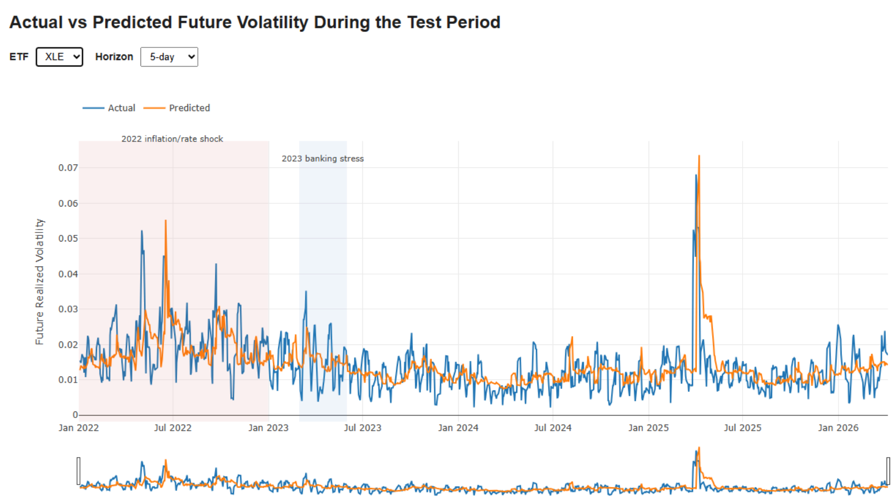


For the 5 day target, every ETF’s best model includes VIX, this is the most concise finding in the project. The largest improvement is in Energy (XLE), where the RF model using baseline + VIX improves RMSE by 5.67% relative to the RF baseline. VIX also improves performance for SPY (3.28%), XLF (3.41%), XLI (3.14%), and XLK (2.68%), with Healthcare-XLV (0.42%) being the least responsive sector.

For the 21 day target (See interactive version), the results are again more multifaceted. SPY, XLI, and XLV are best predicted by the baseline model using only market history. While Energy-XLE remains the most sensitive to uncertainty, with the best performing model being LR with VIX, reducing RMSE by 7.32%. Technology-XLK also shows improvement, with the best model being RF with EPU + EMEU, reducing RMSE by 5.98%.

These results suggest that uncertainty signals matter more consistently across sectors in short horizon forecasting, while providing value more selectively in long horizon forecasting. Furthermore, we again see that news based uncertainty signals provide more value when used in non-linear model classes. This also echoes the findings in the midterm that Energy is the sector most sensitive to uncertainty signals while Healthcare is the opposite.

Figure 3 Prediction Explorer (Interactive)



We can see that even the best models only capture broad volatility patterns, but come short in correctly predicting volatility spikes. This is expected as huge jumps in volatility are often driven by shocks or crisis events that are difficult to forecast with information available up to day t. Predicted volatility is usually smoother than actual volatility, especially for the 21 day target (See interactive version) since volatility is computed over a longer window. Overall, the models are better at identifying broad periods of elevated or reduced volatility, rather than exact timing and size of volatility spikes. This suggests that while uncertainty signals are useful for improving more broad volatility conditions, they are not as consistently useful in supporting the forecast of abrupt movements in the market.

Aggregated RF feature importance was used as a diagnostic to summarise which feature families the non-linear model relied on in the full model. For the 5 day target, we have market-history at 82.6%, VIX at 10.1%, EPU at 3.5%, EMEU at 2.7%, and ticker indicators at 1.1%. For the 21 day target, we have market-history at 77.2%, VIX for 11.9%, EPU for 6.3%, EMEU for 2.8%, and ticker indicators for 1.8%. These importance values confirm the broad pattern of how recent market behavior dominates volatility forecasting, with VIX being the strongest uncertainty signal, followed by EPU, which is more relevant in the long horizon, then EMEU. For LR, we also see this pattern of how market history dominates with VIX being the strongest uncertainty signal.

4. Conclusion and Summary

This project examined whether the uncertainty signals of VIX, EPU and EMEU improve future sector ETF volatility beyond what can be predicted from past market behaviour. Using daily ETF data for SPY, XLE, XLF, XLI, XLK, XLV, we compared the three uncertainty signal families. We forecasted over two horizons, 5 trading days and 21 trading days. The same feature set grid was applied across three models, LR, RF and XGB, and performance was evaluated using test RMSE, test MAE and test R^2.

The main finding is that recent market behaviour consistently remains the strongest predictor of future volatility. Among the uncertainty signals, VIX is the most consistent predictor, improving 5 day forecasts across all 6 sector ETFs, with the largest improvement in Energy. This suggests that market implied volatility is especially useful for near term volatility forecasting. EPU has more limited value, improving performance in non linear models at the 21 day horizon.EMEU does not consistently improve forecasts.

Overall, no single model class dominates all settings. LR performs strongly on the 21 day target when VIX is used, suggesting that in the longer horizon, the relationship between VIX and volatility is captured well linearly. XGB performs the best on the 5 day target when VIX is used, suggesting that on the short horizon, the same relationship may be better captured non-linearly. RF performs well with news based uncertainty signals in the long horizon, suggesting perhaps that news based uncertainty has a non-linear relationship to long horizon volatility.

Sector level results show that Energy is the most uncertainty sensitive sector, while healthcare is the least. Our results indicate that the relationship between uncertainty signals and volatility is a complex one, that depends on the signal type, the forecasting horizon, the model and the sector.

There are several limitations. VIX is not sector specific but market wide, which does not provide sector specific information for more concise market implied volatility reflection. Volatility is calculated from daily close returns rather than high frequency intraday data. This limitation in time resolution is a problem especially if uncertainty signals are priced in much more rapidly than the time horizons we have suggested. Our engineered features are correlated, which prevents individual coefficients or feature importances to be interpreted as separate effects. Furthermore, the test period of 2022-2026 may reflect a period of “outlying” market conditions, marked by inflation, post pandemic adjustment, and geopolitical uncertainty.

Future work could add sector specific predictors such as oil price volatility for Energy or interest rate uncertainty for Financials to explore if there are better signals of market implied uncertainty that can target performance in different sectors. Another extension would be to use intraday data to match the rapid speed of how prices move to news that signals uncertainty. Overall, this project improved on the midterm by investigating more dimensions of uncertainty signals, moving from monthly to daily data frequency, showing that uncertainty matters, but its value is selective instead of universal.